

ASSIGNMENT WEEK-5

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!pip install -q kaggle

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import os
import zipfile

zip_path = "/content/house-prices-advanced-regression-techniques.zip"

with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall()

train = pd.read_csv('train.csv')
test = pd.read_csv('test.csv')

print("Train shape:", train.shape)
print("Test shape:", test.shape)

train.head()

# Combine train and test (excluding SalePrice)
data = pd.concat([train.drop('SalePrice', axis=1), test], axis=0, ignore_index=True)

# Show missing value counts
missing = data.isnull().sum()
missing = missing[missing > 0].sort_values(ascending=False)
print("Missing Values:\n", missing)

# Fill 'None' for NA categorical features
for col in ['PoolQC', 'MiscFeature', 'Alley', 'Fence', 'FireplaceQu']:
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data[col] = data[col].fillna('None')

# Garage-related features
for col in ['GarageType', 'GarageFinish', 'GarageQual', 'GarageCond']:
    data[col] = data[col].fillna('None')
data['GarageYrBlt'] = data['GarageYrBlt'].fillna(0)
data['GarageCars'] = data['GarageCars'].fillna(0)
data['GarageArea'] = data['GarageArea'].fillna(0)

# Basement features
for col in ['BsmtFinSF1', 'BsmtFinSF2', 'BsmtUnfSF',
            'TotalBsmtSF', 'BsmtFullBath', 'BsmtHalfBath',
            'BsmtQual', 'BsmtCond', 'BsmtExposure', 'BsmtFinType1', 'BsmtFinType2']:
    data[col] = data[col].fillna('None' if data[col].dtype == 'object' else 0)

# Mode for categorical, median for numeric
categorical_cols = data.select_dtypes(include='object').columns
numerical_cols = data.select_dtypes(include=['int64', 'float64']).columns

for col in categorical_cols:
    data[col] = data[col].fillna(data[col].mode()[0])

for col in numerical_cols:
    data[col] = data[col].fillna(data[col].median())

print("Missing values after cleaning:", data.isnull().sum().sum())

# Total square footage
data['TotalSF'] = data['TotalBsmtSF'] + data['1stFlrSF'] + data['2ndFlrSF']

# Total bathrooms
data['TotalBath'] = data['FullBath'] + 0.5 * data['HalfBath'] + \

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data['BsmtFullBath'] + 0.5 * data['BsmtHalfBath']
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# Age features
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data['HouseAge'] = 2024 - data['YearBuilt']
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data['RemodAge'] = 2024 - data['YearRemodAdd']
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data['GarageAge'] = 2024 - data['GarageYrBlt']
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# Binary feature: remodeled or not
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data['IsRemodeled'] = (data['YearBuilt'] != data['YearRemodAdd']).astype(int)
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# Simplified quality
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data['SimplOverallQual'] = data['OverallQual'].apply(lambda x: 1 if x < 5 else (2 if x < 7 else 3))
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# One-hot encode
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data = pd.get_dummies(data)
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# Re-split into train/test
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X_train = data.iloc[:train.shape[0], :]
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X_test = data.iloc[train.shape[0]:, :]
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y_train = train['SalePrice']
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print("X_train shape:", X_train.shape)
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```
print("X_test shape:", X_test.shape)
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OUTPUT:

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Train shape: (1460, 81)
Test shape: (1459, 80)
Missing Values:
PoolQC      2909
MiscFeature 2814
Alley       2721
Fence       2348
MasVnrType  1766
FireplaceQu 1420
LotFrontage 486
GarageQual   159
GarageYrBlt  159
GarageCond   159
GarageFinish 159
GarageType   157
BsmtExposure 82
BsmtCond     82
BsmtQual     81
BsmtFinType2 80
BsmtFinType1 79
MasVnrArea   23
MSZoning     4
BsmtFullBath 2
Functional   2
BsmtHalfBath 2
Utilities    2
BsmtFinSF1   1
Exterior2nd  1
Exterior1st  1
Electrical    1
TotalBsmtSF  1
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    BsmtUnfSF      1
    BsmtFinSF2     1
    KitchenQual    1
    GarageArea     1
    GarageCars     1
    SaleType       1
    dtype: int64
Missing values after cleaning: 0
X_train shape: (1460, 309)
X_test shape: (1459, 309)
```